Hostel Management System

Branch Coverage Report

Final regression run, 11 November 2026. Team: Krish Mathur, Atharva Mihir Gore, Samyuktha, Tanishk Gowda.

1. Summary

Run date	11 November 2026
Code under test	feature/part3-testing (based on main at c7f9932 plus feature/student-home)
Environment	Windows 11, Python 3.11.9, Django 5.2.19, SQLite 3, coverage.py 7.17.2
Command	coverage run --branch --source=hostels,rooms,allocations,dashboard,accounts,students manage.py test; coverage report -m; coverage json
Application code	127 of 128 branches covered (99.2%), 1 partial
Including test files	216 of 218 branches covered (99.1%)
CI	The GitHub Actions workflow runs coverage with --branch on every push and pull request to main

Branch coverage counts each possible jump at an if, loop or try: for example both the true and false outcome of an if. A **partial** branch is a line where at least one possible jump was never taken.

2. Branch coverage by module (application code)

Module	Branches	Covered	Missed	Coverage
accounts	52	51	1	98.1%
allocations	34	34	0	100.0%
dashboard	0	0	0	no branches
hostels	6	6	0	100.0%
rooms	18	18	0	100.0%
students	18	18	0	100.0%
Total	**128**	**127**	**1**	**99.2%**

3. Branch coverage by file (application code, files with branches)

File	Branches	Covered	Partial	Coverage
accounts/decorators.py	8	8	0	100%
accounts/forms.py	10	10	0	100%
accounts/services.py	10	9	1	90%
accounts/validators.py	18	18	0	100%
accounts/views.py	6	6	0	100%
allocations/forms.py	2	2	0	100%
allocations/models.py	2	2	0	100%
allocations/services.py	18	18	0	100%
allocations/views.py	12	12	0	100%
hostels/forms.py	2	2	0	100%
hostels/views.py	4	4	0	100%
rooms/models.py	6	6	0	100%
rooms/views.py	12	12	0	100%
students/forms.py	8	8	0	100%
students/services.py	6	6	0	100%
students/views.py	4	4	0	100%

4. Partial branches explained

File	Jump not taken	Explanation
accounts/services.py	line 36 to exit	In logout_user(), the case where the request has no session attribute. Every real request passes through SessionMiddleware, so this defensive guard is never false in tests. Not a defect.
allocations/tests.py	line 278 to line 290	In UT_ALLOC_05, the retry loop running all 50 attempts without success. The test always finishes in a few attempts. Test code only.